

HORSESHOE BAY, TX, USA

WMO: 720639

Lat: 30.533N

Lon: 98.367W

Elev: 333

StdP: 97.39

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.0	-11.2	1.5	3.3	-8.7	1.9	4.4	8.7	12.8	7.6	12.0	2.1	0	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	38.2	21.9	37.4	22.0	36.3	22.2	24.7	30.5	24.3	30.3	24.0	30.0	3.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.2	25.7	22.9	18.4	25.9	22.6	18.0	25.8	76.7	30.9	75.1	30.4	73.7	29.4	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.5	5.6	-7.1	40.1	2.8	1.7	-9.1	41.3	-10.7	42.3	-12.3	43.3	-14.4	44.6		
			-8.4	25.6	2.2	0.6	-10.0	26.1	-11.3	26.5	-12.5	26.8	-14.1	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	9.7	12.2	16.5	20.3	23.9	27.9	29.3	29.8	26.2	20.8	14.9	10.6
	DBStd	8.22	5.19	6.14	5.07	4.04	3.70	2.00	1.76	2.06	3.00	4.62	5.38	5.15
	HDD10.0	197	69	44	9	0	0	0	0	0	1	18	56	56
	HDD18.3	986	272	184	96	26	6	0	0	0	1	29	128	244
	CDD10.0	3925	58	105	210	309	431	538	599	613	486	336	165	74
	CDD18.3	1671	2	12	37	86	179	288	341	354	237	106	24	5
	CDH23.3	19219	37	96	313	769	1704	3409	4418	4735	2531	1004	159	44
	CDH26.7	9748	3	19	72	245	713	1764	2506	2769	1261	359	32	4
(14) Wind	WSAvg	2.3	2.1	2.4	2.7	2.9	2.8	2.7	2.3	2.1	1.8	2.1	2.1	1.9
(15) Precipitation	PrecAvg	778	45	46	67	63	96	82	53	49	72	82	67	46
	PrecMax	1197	118	162	211	165	199	296	242	251	185	267	251	248
	PrecMin	332	0	0	0	7	21	16	0	1	3	0	0	0
	PrecStd	219	34	42	43	42	49	67	54	53	47	68	59	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	29.3	30.7	33.2	36.2	38.8	39.8	40.1	37.8	33.9	30.1	27.4
		MCWB	15.1	15.7	17.7	19.8	22.4	21.9	21.4	21.9	20.4	20.7	20.1	16.7
	2%	DB	23.9	26.2	28.7	31.3	33.8	37.0	37.9	38.6	36.1	32.2	27.2	24.3
		MCWB	14.7	15.9	18.2	19.9	21.7	22.1	21.8	22.2	21.3	21.7	18.7	16.9
	5%	DB	21.4	23.8	27.0	29.5	32.3	35.2	37.1	37.5	34.5	30.4	25.0	22.1
		MCWB	14.1	16.3	17.2	19.3	21.5	22.4	22.0	22.4	21.5	20.5	18.0	16.3
	10%	DB	18.9	21.8	24.9	27.7	30.9	33.9	35.9	36.5	32.9	28.7	23.1	19.9
		MCWB	12.8	16.1	16.6	18.5	21.6	22.5	22.3	22.3	21.6	19.5	17.9	15.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	20.4	20.7	23.4	24.7	25.3	25.2	24.9	25.0	24.4	22.6	20.6
		MCDB	21.6	24.1	25.7	28.4	29.8	31.4	30.4	32.2	30.0	28.6	26.5	23.7
	2%	WB	17.6	19.5	20.0	22.2	23.9	24.6	24.6	24.3	24.2	23.5	21.5	19.3
		MCDB	20.9	22.4	24.4	27.5	29.0	30.7	30.2	31.5	29.5	27.8	24.5	21.9
	5%	WB	15.8	18.4	19.3	21.4	23.3	24.1	24.1	24.0	23.8	22.7	20.2	17.8
		MCDB	19.2	21.2	23.6	26.5	28.6	30.2	30.1	31.1	29.0	27.1	23.1	20.4
	10%	WB	14.0	17.1	18.6	20.6	22.7	23.7	23.8	23.6	23.4	21.9	19.0	15.9
		MCDB	18.3	20.2	23.0	25.3	28.1	29.9	29.9	30.8	28.3	26.2	22.0	18.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	12.6	13.2	12.7	11.5	11.9	12.6	13.0	12.1	12.9	12.6	12.2
		MCDBR	17.3	14.9	14.7	14.3	13.6	13.7	14.4	14.0	14.1	13.7	13.4	15.4
	5% WB	MCWBR	9.3	7.9	6.8	6.2	4.5	2.9	2.9	3.0	3.8	5.2	7.0	8.9
		MCWBR	12.8	11.1	11.7	11.0	10.2	10.8	11.6	12.4	10.4	10.4	10.1	12.0
		MCWBR	8.7	7.7	6.3	5.8	4.2	2.9	2.8	2.9	3.2	5.1	6.7	9.2
(40) Clear-Sky Solar Irradiance	taub	0.315	0.322	0.349	0.389	0.424	0.417	0.428	0.419	0.396	0.366	0.343	0.315	0.315
	taud	2.524	2.509	2.444	2.353	2.308	2.372	2.382	2.387	2.403	2.470	2.470	2.527	2.527
	Ebn at Noon	921	945	936	904	869	870	859	865	876	883	878	901	901
	Edh at Noon	87	96	110	125	132	123	121	119	113	99	91	83	83
(44) All-Sky Solar Radiation	RadAvg	3.11	3.67	4.66	5.57	5.85	6.74	6.74	6.36	5.21	4.27	3.25	2.76	2.76
	RadStd	0.34	0.57	0.43	0.52	0.60	0.48	0.55	0.35	0.49	0.62	0.41	0.37	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page